Business Intelligence Overview

Business intelligence, executive dashboards and decision support | Superstore retail case study

01 Executive context

Business Intelligence (BI) converts operational data into trusted information that managers can use to monitor performance, diagnose problems and choose actions. It combines data acquisition, preparation, modelling, metrics, visualisation and governance. The objective is not to produce charts for their own sake, but to shorten the distance between a business question and a defensible decision [1, 2].

02 Why organisations use dashboards

Executives need consistent answers to recurring questions: Are sales growing? Where is profit concentrated? Which customer segments perform well? Are revenue gains translating into earnings? A dashboard consolidates decision-critical measures and highlights changes that would be difficult to see in raw transaction files.

- Single performance view: standard KPI definitions limit conflicting departmental figures.
- Faster detection: trends, comparisons and exceptions surface emerging issues.
- Self-service analysis: filters enable management to investigate a region, product or period.
- Accountability: named metrics allow decisions and interventions to be tracked over time.
- Communication: visual hierarchy helps non-specialists interpret complex datasets.

03 How dashboards improve decisions

Management question	Dashboard evidence	Potential decision
Are sales improving?	Monthly trend; YoY sales growth	Adjust demand planning and commercial priorities
Which markets underperform?	Sales and margin by region/state	Review product mix, pricing and local execution
What contributes most to earnings?	Profit by category and product	Prioritise high-return assortments
Are discounts sustainable?	Margin by discount band	Review approvals and promotion rules

Interactive dashboards add value when slicers, drill-downs and tooltips preserve a single source of metric definitions. Their usefulness depends on reliable data and a clear distinction between descriptive patterns and causal claims [1, 4].

04 Scope of the selected dataset

The selected Sample Superstore dataset contains 9,994 US retail order-line records spanning 2014–2017 (by Order Date), including Sales, Profit, Quantity, Discount, Product, Customer, Segment and Region. Each row represents an order line, not a whole order. Accordingly, the number of orders must be calculated as distinct Order IDs. The dataset has no inventory, operating-expense detail or customer-acquisition cost information.

05 Business objective and decision questions

The consulting brief requires a Power BI executive report to monitor sales performance, profitability, customer behaviour and regional contributions, and to recommend evidence-based improvements. The case-study decision objective is to determine where the retail business earns value, where it loses value and what management should investigate first.

Decision area	Primary measures / views
Enterprise performance	Total Sales; Total Profit; Total Orders; Average Sales per line; Profit Margin
Geography	Regional and state-level sales/profit; geographic distribution
Customers	Sales and margins by Consumer, Corporate and Home Office segments
Products	Category, sub-category and product profitability
Time	Monthly sales movement and annual profitability

06 Proposed BI delivery workflow

1 Acquire and profile. Import the source CSV; inspect grain, completeness, types and exceptions.

2 Transform and validate. Use Power Query to standardise types and text; preserve legitimate loss-making records.

3 Model and measure. Relate a unique Date dimension to order-line facts; build explicit DAX measures.

4 Visualise and test. Create KPI cards and required visuals; verify totals and filter interactions.

5 Interpret and recommend. Quantify insights, qualify limitations and set monitoring indicators.

07 Controls, limitations and responsible use

Reporting quality depends on explicit business rules: aggregate Sales and Profit over line items; count distinct Order IDs; calculate Profit Margin as total profit divided by total sales; and use Order Date as the default time basis. Product IDs are not consistently one-to-one with names, so product-name grouping needs care. The available data is historical, transactional and observational: it supports descriptive diagnosis, but not forecasts or causal claims without further investigation.

08 Sources and reference material

[1] Microsoft (n.d.). What is business intelligence? https://www.microsoft.com/en/power-platform/products/power-bi/topics/business-intelligence/what-is-business-intelligence

[2] IBM (n.d.). What is business intelligence? https://www.ibm.com/think/topics/business-intelligence

[3] Microsoft Learn (n.d.). Understand star schema and the importance for Power BI. https://learn.microsoft.com/en-us/power-bi/guidance/star-schema

[4] Microsoft Learn (n.d.). Design Power BI reports for accessibility. https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-accessibility-creating-reports

Project data: Sample - Superstore.csv, supplied with the AnalystLab Africa Week 2 assignment. Public reference pages accessed 28 October 2026.